Rectal adenocarcinoma

Neoadjuvant chemoradiation / TNT distinguishes rectal from colon management; staging, prognosis and metastatic systemic therapy are as for colon cancer

Stanford Oncology Handbook 2025, Rectal Cancer (Tam AY, Shaheen S) · NCCN · items marked † are outside the handbook chapter

WORKUP

- Colonoscopy + biopsy with **MMR / MSI testing** and pathology review
- **MRI pelvis (3T)** — the rectal-specific study; rectal ultrasound if MRI contraindicated, inconclusive, or for superficial lesions
- CT chest/abdomen/pelvis · FDG-PET if potentially curable M1
- Labs: CBC, CMP, **CEA**
- **PIK3CA testing in stage II–III** — if mutated, aspirin 100–162 mg daily × 3 years post-op
- If metastatic: KRAS, BRAF, HER2, POLE/POLD1, RET, NTRK
- Genetics referral for all rectal cancer **<50 y**, or a somatic finding with germline implications

RISK DEFINITIONS

High-risk rectal	cT4a or cT4b · **extramural vascular invasion** · cN2 · **involved mesorectal fascia** · enlarged lateral nodes considered metastatic
High-risk path after local excision	positive margins · lymphovascular invasion · poorly differentiated · submucosal invasion to the lower third
T4 (rectal nuance)	direct extension through serosa into prostate, seminal vesicles, cervix or vagina

TNM is otherwise identical to colon — see the colon one-pager for stage groups and for all metastatic systemic therapy.

KEY TRIALS

PROSPECT	RT-sparing: FOLFOX/CAPEOX × 12–16 wk with radiation reserved for inadequate regression
RAPIDO	short-course RT + chemo TNT: pCR **28% vs 14%**; lower distant relapse but **possible higher local recurrence** (Dijkstra, Ann Surg 2023)
PRODIGE 23	induction mFOLFIRINOX: pCR **27.8% vs 12.1%** (Conroy)
OPRA	watch-and-wait after clinical CR to TNT preserves the organ without compromising survival
Cercek (NEJM 2025)	dostarlimab in **dMMR** rectal cancer — basis for immunotherapy over TNT
SHORTFOX	ongoing — RT omission and course length

> **Watch-and-wait requires a complete clinical response on all three of DRE, rectal MRI, and direct endoscopic evaluation** — and only in centers with experienced multidisciplinary teams, after careful discussion with the patient.
>
> **RAPIDO caveat:** short-course RT lowered distant relapse but may raise local recurrence, so high-risk patients get TNT as induction chemo followed by chemoradiation. Short-course candidacy is a multidisciplinary decision weighing downstaging against long-term toxicity.

TREATMENT PATHWAY

Early — cT1–2 N0

cT1 N0
endoscopic submucosal dissection, or
transanal local excision
pT1 without high-risk features → surveillance

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pT1 high-risk or pT2
transabdominal resection preferred
then 6 months adjuvant chemoRT + chemo
or TNT, then restage → surgery

Locally advanced — pMMR / MSS

TNT is the standard
FOLFOX or CAPEOX + chemoRT
concurrent capecitabine during radiation
~6 months of total neoadjuvant therapy

↓

High-risk disease sequencing
induction chemo THEN chemoRT
driven by the RAPIDO local-recurrence signal
induction mFOLFIRINOX an option (PRODIGE 23)

↓

RT-sparing (PROSPECT-eligible)
non-T4, ≤4 nodes, sphincter-sparing candidate
FOLFOX/CAPEOX 12–16 wk → restage
add RT only if regression inadequate

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After neoadjuvant therapy
restage → transabdominal resection
if clinical CR: **watch and wait** (see note)

dMMR / MSI-H or POLE-POLD1

Immunotherapy, preferred OVER TNT
dostarlimab, nivolumab, or pembrolizumab
up to 6 months; reassess every 2–3 months
(Cercek, NEJM 2025)

↓

Then
clinical CR → surveillance, **no surgery**
persistent at 6 mo → chemoRT → surgery

REGIMENS & DOSING

Regimen	Dosing	Schedule
ChemoRT (infusional 5-FU)	5-FU **225 mg/m²** IV over 24 h, 5 days per week	with RT × 5 wk
ChemoRT (capecitabine)	capecitabine **825 mg/m² PO BID** on days of radiation	concurrent with RT
Long-course radiation	5–6 weeks	—
Short-course radiation	5 days	—
Dostarlimab	500 mg IV × 4 doses, then 1000 mg IV	q3wk → q6wk
Nivolumab	240 mg (or 3 mg/kg if <40 kg) · or 480 mg	q2wk · q4wk
Nivolumab + ipilimumab	nivo 3 mg/kg + ipi 1 mg/kg × 4, then nivo maintenance	q3wk → q2–4wk
Aspirin (PIK3CA-mutant)	100–162 mg PO daily × 3 years post-op	stage II–III
FOLFOX · CAPEOX · mFOLFIRINOX	named only here — dosing on the colon page	—

BIOMARKER → THERAPY

Alteration	Therapy
dMMR / MSI-H or POLE-POLD1 (TMB >50)	Neoadjuvant/definitive immunotherapy up to 6 months — **preferred over TNT**; dostarlimab, nivolumab or pembrolizumab
Same, metastatic	ICI preferred if no prior exposure: nivolumab ± ipilimumab, pembrolizumab, dostarlimab, cemiplimab, retifanlimab, toripalimab, tislelizumab
pMMR/MSS, TMB ≥10	Pembrolizumab in the refractory setting (KEYNOTE-158)
PIK3CA, stage II–III	Aspirin 100–162 mg daily × 3 years after surgery
ctDNA	Prognostic only — **not** for treatment decisions outside a trial

SURVEILLANCE

- Same as colon cancer for stage I–IV, **except after transanal local excision**
- **Transanal local excision:** proctoscopy (with EUS or MRI) **q3–6 months × 2 years**, then q6 months to 5 years total
- Colonoscopy at **1 year**; if advanced adenoma repeat in 1 year, otherwise 3 years then every 5 years
- dMMR on immunotherapy: re-evaluate **q2–3 months**; MSS metastatic being converted to resectable: **q2 months**

PEARLS

- **FOLFOXIRI is not recommended** when resection is contraindicated after neoadjuvant therapy
- Resection is **preferred over ablation or SBRT** for oligometastatic liver- or lung-only disease
- Favorable pT3 N0 (well/moderately differentiated, <2 mm mesorectal invasion, no lymphovascular involvement, upper rectum) may go to surveillance rather than adjuvant therapy
- pT1–3 N1 can be treated with FOLFOX or CAPEOX alone, without radiation